

CATHERINE CHENG

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

B.S. Expected May 2028

Bachelor of Science, Electrical and Computer Engineering

Relevant Coursework: Computer Architecture, Computer Systems Programming, Digital Logic, Digital System Design - Microcontrollers, Electronic Circuits, Embedded Systems, Electromagnetism, Data Science, Differential Equations, Linear Algebra, Microelectronics, Power Electronics, Signals and Systems

TECHNICAL SKILLS

- **Programming & HDLs:** C/C++, Python, Perl, SystemVerilog/Verilog, Bash/Linux, Makefile, CMake
- **Embedded & Protocols:** Embedded C/C++, Arduino, Raspberry Pi, UART, I2C, PWM, PID Control, State Machines
- **Hardware & Equipment:** RTL Design, FPGA, PCB/Avionics Wiring, Oscilloscope, DMM, Signal Generator, Power Supply
- **Software & Tools:** LTspice, PLECS, Intel Quartus Prime, ModelSim, Verilator, Git/GitHub, AutoCAD, Valor, Betaflight

PROJECTS

FPGA TinyRV1 Processor

August 2025 – December 2025

Digital Logic

Ithaca, NY

- Designed and synthesized a single-cycle TinyRV1 processor in SystemVerilog on a Terasic DE0-CV FPGA, implementing the complete processor datapath and control logic
- Implemented modular RTL components including 32-bit adders, a 32-register register file, multiplexers, and input synchronizers to construct the processor datapath
- Verified processor functionality through directed, randomized, and X-propagation testing, achieving full instruction-set coverage and resolving RTL bugs across edge cases

UART Number Guessing

April 2026 – May 2026

Embedded Systems

Ithaca, NY

- Developed a two-board embedded system on FRDM-KL46Z ARM Cortex-M0+ microcontrollers, implementing UART communication and interrupt service routines (ISRs) to synchronize game state and user inputs in real time
- Implemented mutexes, semaphores, and spinlocks to coordinate concurrent tasks and safely manage shared resources within the embedded system
- Integrated LCD displays, GPIO buttons, and PWM controlled LEDs with interrupt driven control to provide responsive user interaction and real-time game feedback

EXPERIENCE

Northrop Grumman

June 2026 – August 2026

Computer Engineer Intern

Baltimore, Maryland

- Developed a Perl-based tool to automate, verify, and standardize PWB design data processing, reducing a 25+ minute manual workflow to 4 minutes while limiting engineer involvement to final inspection
- Revaluated outdated dielectric withstand voltage (DWV) requirements through hardware testing and analysis, supporting a safe reduction in test voltage to reduce unnecessary PWB stress and improve manufacturing yield
- Automated component-selection verification, reducing verification time 90% from 2 hours to 12 minutes and establishing the workflow as the Power group's standard for parts orders

Autonomous Drones Project Team

August 2025 – present

Electrical Subteam

Ithaca, NY

- Debugged power-distribution issues on IARC competition drones where components were not receiving sufficient current, redesigning electrical wiring to improve power delivery while reducing wiring complexity
- Integrated GPS and LiDAR sensors over UART and I2C to provide navigation and ranging data for autonomous flight
- Improved quadcopter flight stability through systematic PID tuning and IMU calibration using Betaflight and Pixhawk flight controllers

Entertainment Engineering and Design

January 2025 – May 2026

Electrical Subteam

Ithaca, NY

- Built motorized Flappy Bird game using stepper motor and driver circuit to control obstacle movement timing
- Assembled and soldered motion control electronics for animatronic arm mechanism, integrating servo motors for coordinated movement
- Developed Arduino C++ PID velocity-control algorithms to achieve repeatable motor motion and coordinated movement sequences

Duffield Engineering Office

August 2025 – current

Peer Advisor

Ithaca, NY

- Led weekly class sections for ~50 first-year engineering students across two sections, providing academic guidance and serving as a liaison between students and engineering administration
- Designed and led interactive activities and organized academic and social events, including off-campus educational trips to museums, to strengthen student engagement and community